

SURAJ PADALA

📞 +91-9392801348 | ✉ padalasuraj80@gmail.com | 🌐 padala-suraj | 🌐 padalasuraj

PROFESSIONAL SUMMARY

Application Developer at Cognizant with hands-on experience building secure REST APIs, data integration pipelines and backend workflows. Graduate from Army Institute of Technology Pune, skilled in full-stack development and scalable application design. Currently expanding expertise in cloud computing and DevOps technologies with a strong interest in modern software engineering practices.

TECHNICAL SKILLS

Languages:	Python, C++ , JavaScript (ES6+), SQL, Java(learning)
Frameworks:	React.js, Node.js, Express.js, Angular, Bootstrap, HTML5, CSS3
Databases:	MongoDB, MySQL, HSQL
DevOps & Tools:	Git, GitHub Actions, Docker, CI/CD Pipelines, Firebase, Postman, VS Code
Platforms:	Mendix (Rapid & Intermediate Certified), REST APIs, OData, Swagger UI
Concepts:	DSA, OOP's, RBAC, SSO, Microflows

EXPERIENCE

Cognizant Sep 2025 – Present
Application Developer — Full Time Pune, India

- Developed API-based data integration workflows and secure **RESTful APIs** with **API key authentication** for the **New York TRS Project** using **REST and OData services**, reducing data mismatches by **40%** across multiple systems
- Implemented **Role-Based Access Control (RBAC)** and **SSO integration** to strengthen access control and managed API design, testing, and documentation using **Swagger UI**
- Developed **microflows** and **Java actions** to streamline application workflows, while optimizing **SQL/HSQL queries** to improve performance and maintain data integrity
- Performed **log analysis**, **root-cause debugging**, and **structured error handling** to improve application stability and reduce production issues in an Agile development environment

PUBLICATION

Hand Gesture Controlled Robot Using STM32 Cortex-M4 and MPU6050 [🔗](#) IEEE Xplore 2025
IEEE — ICEC2NT 2025

- Developed firmware for a hand gesture controlled robot using **STM32 Cortex-M4** and **MPU6050** to process real-time accelerometer and gyroscope data
- Implemented sensor fusion and motion control logic in **STM32CubeIDE**, achieving **99.8% gesture recognition accuracy** for robotic movement control
- Optimized **BLE communication**, PWM motor control, and interrupt-driven execution using **DMA**, improving response time and reducing CPU load during real-time operation

PROJECTS

AI-Based Q&A System for Video Content with Multimodal Integration 2025
RAG, Natural Language Processing

- Developed a **multimodal pipeline** using **Natural Language Processing** techniques to enable interactive querying of video content
- Utilized a **Retrieval-Augmented Generation (RAG)** model for searching transcribed text and extracting relevant information
- Integrated audio and text processing to transform video content into an **interactive** and **accessible knowledge base**
- Utilized a pretrained **Llama 3** model to generate **detailed** and **context-aware answers** from the extracted information

Campus Club Website — AIT E-Cell [🔗](#) 2023 – 2024
React, Node.js, Express, MongoDB, REST APIs — Full-Stack

- Built and maintained the official **E-Cell AIT** platform using the **MERN Stack**, supporting event registrations and content management for **500+ students**
- Integrated **RESTful APIs** for registrations and event updates, and developed responsive UI components for seamless usage across desktop and mobile devices

BST Visualizer Pro [🔗](#) 2023
JavaScript, D3.js — Web Tool

- Built a **Binary Search Tree visualizer** using **D3.js** and **SVG**, supporting real-time node insertion, deletion, search, and traversal animations with adjustable speed controls
- Developed a responsive user interface with **Bootstrap** and **JavaScript**, deployed on **Netlify** with clean **HTML / CSS** for consistent performance across browsers

EDUCATION

Army Institute of Technology, Pune

B.E., Electronics & Telecommunication Engineering (CGPA: 7.76 / 10)

2021 – 2025

Pune, India

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certified Cloud Practitioner (CLF-C02)** — Foundational knowledge of AWS Cloud services, security and billing
- **Mendix Certified Developer** — Rapid & Intermediate Application Development (Mendix Academy)
- **Full-Stack Web Development** — React, Node.js, RESTful APIs, Bootstrap (Udemy)
- **AIR 294** in Coding Ninjas National GI Contest — top **1%** among **50,000+ participants** (2023)
- Solved **500+ problems** on LeetCode, Codeforces & GFG focusing on optimized algorithm logic